

# Variance-Reduced Q-Learning over Static and Time-Varying Networks

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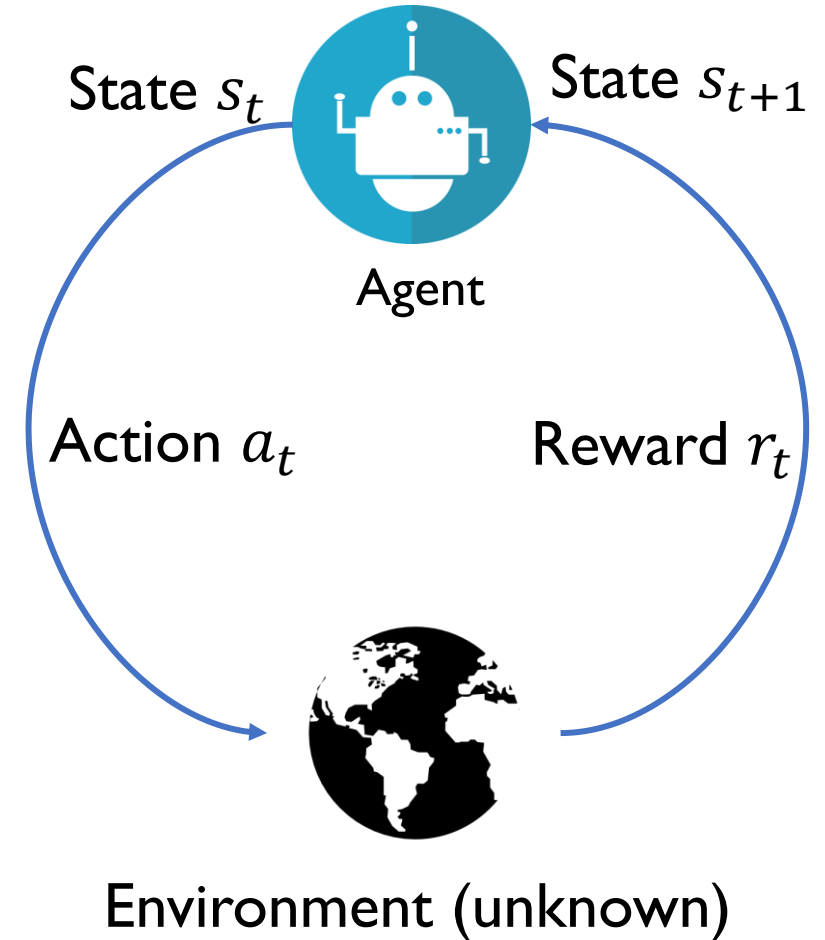
# **Part I: Motivation & Background**

# Reinforcement Learning (RL)

An *online* learning paradigm where agents learn to make decisions by interacting with an **UNKNOWN** environment

**Goal:** Maximize the cumulative reward over time

**Q.** How do agents make “**good**” decisions under *environmental uncertainty*?



# Reinforcement Learning: Applications

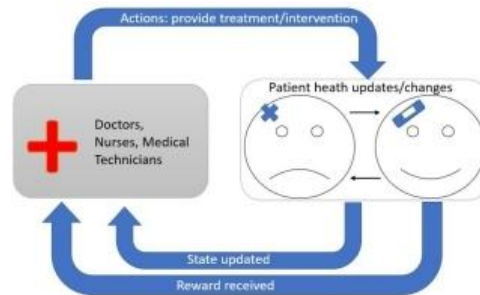


Autonomous driving [1]



Manipulation [2]

**Robotics**



AI prescription [3]

**Healthcare**



Beamforming [4]

**Wireless Communication**

[1] Deep reinforcement learning for autonomous driving: A survey, Kiran, B. R. et al., TIT, 2021.

[2] Review of deep reinforcement learning for robot manipulation, Nguyen, H., & La, H., IRC, 2019.

[3] Reinforcement learning in healthcare: A survey, Yu, C. et al., CSUR, 2021.

[4] Deep reinforcement learning for 5G networks: Joint beamforming, power control, and interference coordination, Mismar, F. B. et al., IEEE Transactions on Communications, 2019.

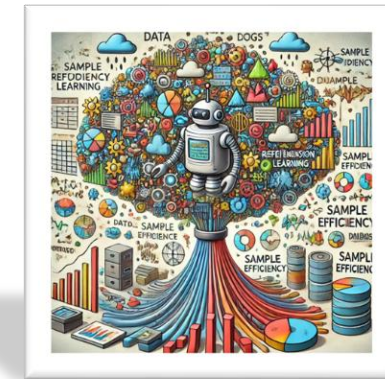
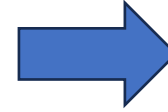
# Open Challenges in RL



**High-dimensional data**



**Massive state and action spaces**

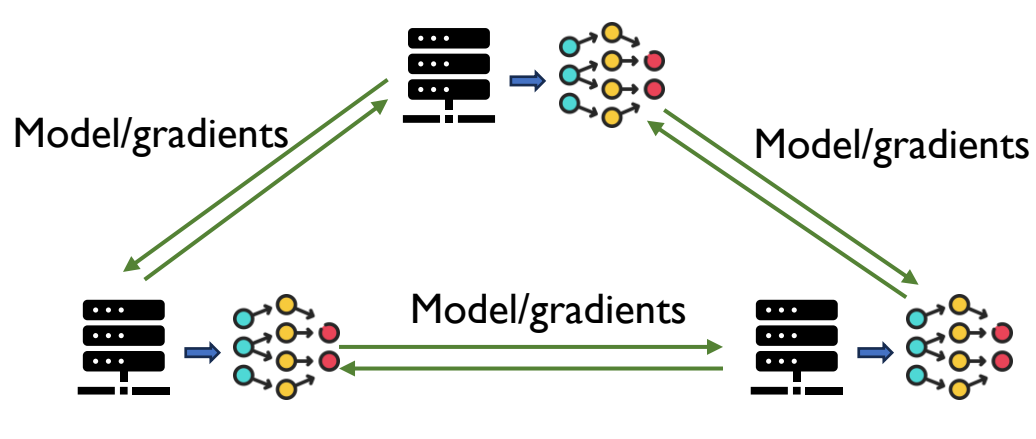


**Data hungry algorithms [1]**

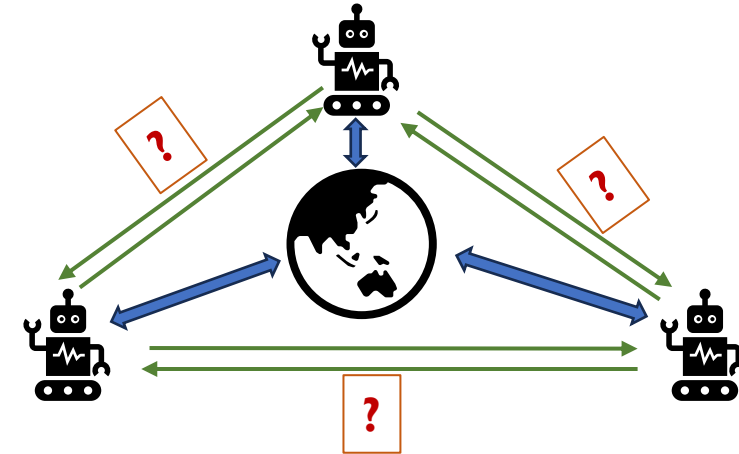
**Setting:** Multiple agents interact with the same environment over a network

**Q. Can agents collaborate by sharing info over a network to learn to make good decisions **faster**?**

# Decentralized Reinforcement Learning



Supervised learning



Online decision-making

## Goals in decentralized RL:

- Learn the **optimal policy** for the environment
- Demonstrate **collaborative speedup** in the **sample-complexity bounds**, i.e., multiple agents do help **expedite** the learning
- Achieve the above with as little **communication** as possible
- Robustness to **time-varying** networks

# Contributions

In this work, we developed an algorithm that **simultaneously** achieves these goals:

- An **optimal** policy at each agent
- **Linear speedup** w.r.t. the number of agents
- **Poly-logarithmic communication**
- Extension to **time-varying** networks

# **Part II: Prior Work**



# Introduction to Q-Learning

**Goal:** Find the optimal policy  $\pi^*$  of environment  $\mathcal{M} := \{\mathcal{S}, \mathcal{A}, R, \mathcal{P}, \gamma\}$

- At **state**  $s_t$ , agent plays **action**  $a_t$ , and transitions to the next **state**  $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$
- **Policy**  $\pi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$ , **action** at **state**  $s_t$  being  $a_t \sim \pi(\cdot | s_t)$
- **Reward** at time-step  $t$ :  $R(s_t, a_t)$ ; assume bounded by  $\bar{R}$
- Fix a policy  $\pi$ , the **state-action value function** (Q-function)  $Q^\pi \in \mathcal{S} \times \mathcal{A}$  is

$$Q^\pi(s, a) = R(s, a) + \mathbb{E} \left[ \sum_{t=1}^{\infty} \gamma^t R(s_t, a_t) \mid \pi \right]$$

# Introduction to Q-Learning (cont.)

**Goal:** Find the optimal policy  $\pi^*$  of environment  $\mathcal{M} := \{\mathcal{S}, \mathcal{A}, R, \mathcal{P}, \gamma\}$

- Find the optimal  $Q^*$  associated with  $\pi^*$ , and extract  $\pi^*$  via

$$\pi^*(s) = \arg \max_a Q^*(s, a)$$

- Environment dynamics  $R, \mathcal{P}$  **unknown**, learn  $Q^*$  from interaction (**Q-learning**):

$$Q_{t+1}(s, a) = (1 - \alpha)Q_t(s, a) + \alpha \mathcal{T}_t Q_t(s, a)$$

$$\mathcal{T}_t f(s, a) = R(s, a) + \gamma \max_{a'} f(s_t(s, a), a'), \quad s_t(s, a) \sim \mathcal{P}(\cdot | s, a)$$

$\square \mathcal{T}_t$ : estimate of the *Bellman optimality operator* at time-step  $t$

# From Single-agent Q to Decentralized Q

**Goal:** Each agent  $i \in [N]$  learns the optimal policy  $\pi^*$  of  $\mathcal{M}$

- All agents in  $[N]$  interacts with the same environment  $\mathcal{M} := \{\mathcal{S}, \mathcal{A}, R, \mathcal{P}, \gamma\}$
- Agents communicate over a *connected, undirected* graph  $\mathcal{G}$
- Agents need to *diffuse* information throughout the network to accelerate learning

**Multiple challenges for decentralized  
Q-learning**

# Challenge 1: Achieving Linear Speedup

- Benchmark rate for **single-agent** Q-learning [1, 2]: w.h.p.,

$$\|Q_T - Q^*\|_\infty \leq \tilde{O}\left(\frac{1}{\sqrt{T}}\right)$$

**Q1.** Since there are  $N$  agents, can we **accelerate the rate** by a factor of  $N$ ?



**Goal:**  $\|Q_T - Q^*\|_\infty \leq \tilde{O}\left(\frac{1}{\sqrt{NT}}\right)$  w.h.p.

[1] *Stochastic approximation with cone-contractive operators: Sharp  $\ell_\infty$ -bounds for Q-learning*, Wainwright, M. J., arXiv, 2019.

[2] *Finite-time analysis of asynchronous stochastic approximation and Q-learning*, Qu, G., & Wierman, A., CLT, 2019.

# Challenge 2: Communication Overhead

| Work      | Linear Speedup | Comm. Overhead           |
|-----------|----------------|--------------------------|
| [1]       | NO             | Linear with $T$          |
| [2]       | NO             | Linear with $T$          |
| [3]       | NO             | Linear with $T$          |
| [4]       | NO             | Linear with $T$          |
| This work | <b>YES</b>     | <b>Poly-log</b> with $T$ |

**Q2.** Can each agent learn an **optimal** policy demonstrating **linear speedup**, with **little communication**?

[1] *Finite-time analysis of distributed TD(0) with linear function approximation on multi-agent reinforcement learning*, Doan T., Maguluri S., Romberg J., ICML, 2019

[2] *Finite-sample analysis of distributed Q-learning for multi-agent networks*, Heredia P., Ghadialy H., Mou S., ACC, 2020

[3] *A finite time analysis of distributed Q-learning*, Lim H., Lee D., arXiv, 2024

[4] *Finite-time convergence rates of decentralized stochastic approximation with applications in multi-agent and multi-task learning*, Zeng S., Doan T., Romberg J., IEEE TAC, 2022

# **Part III: Variance-Reduced Q-Learning (VRDQ)**

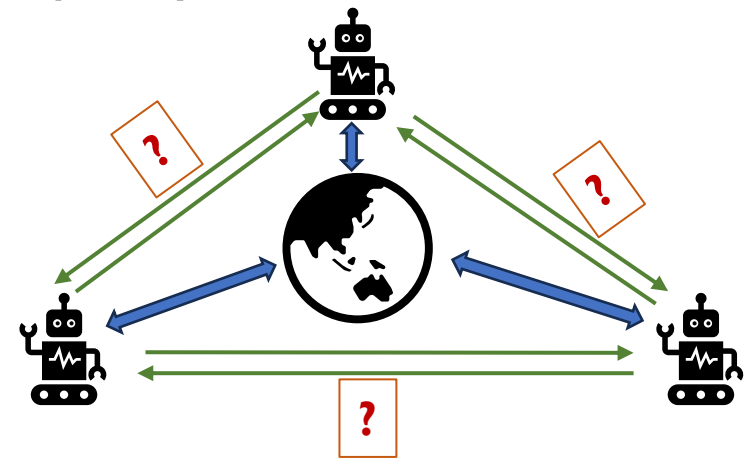
# Fundamental Questions in Decentralized RL

**Goal:** Each agent  $i \in [N]$  learns the optimal policy  $\pi^*$  of  $\mathcal{M}$

- All agents in  $[N]$  interacts with the same environment  $\mathcal{M} := \{\mathcal{S}, \mathcal{A}, R, \mathcal{P}, \gamma\}$
- Agents communicate over a *connected, undirected* graph  $\mathcal{G}$
- Agents need to *diffuse* information throughout the network to accelerate learning
- Consider a **generative synchronous sampling model** for simplicity

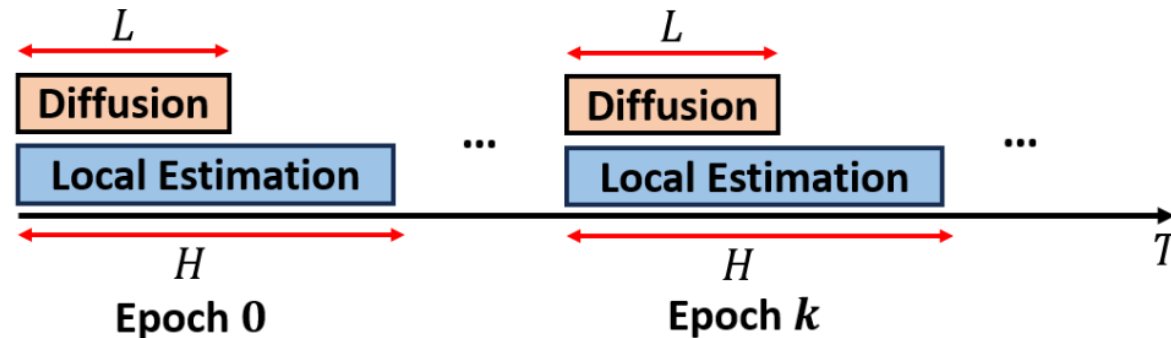
**Q1:** What info should agents diffuse?

**Q2:** How often should such info be diffused?



# VRDQ – Overview

- Two **decoupled** modules
- **Module 1 – Operator estimation**: Obtain a **variance-reduced** estimate of the Bellman optimality operator by dividing samples into epochs
- **Module 2 – Diffusion**: Parallel operation of estimation and diffusion





# VRDQ – Module I: Operator Estimation

**Key idea:** **Refine** the operator with **more samples** and **update** the estimate **fewer times** instead of frequently updating with a noisy operator from **a single sample**.

- Recall Q-learning update rule -  $\mathcal{T}_t$  constructed from a **single**  $(s, a)$  pair:  
$$Q_{t+1}(s, a) = (1 - \alpha)Q_t(s, a) + \alpha \mathcal{T}_t Q_t(s, a)$$
$$\mathcal{T}_t f(s, a) = R(s, a) + \gamma \max_{a'} f(s_t(s, a), a')$$
- Our algo:** Each agent has  $T$  samples; divide into  $K$  epochs with  $H$  samples each ( $T = KH$ ); construct a **refined** operator  $\mathcal{T}_{i,k}$  using  $H$  samples in epoch  $k$ :

$$\mathcal{T}_{i,k} f(s, a) = R(s, a) + \gamma \sum_{s' \in \mathcal{S}} \hat{\mathcal{P}}_{i,k}(s' | s, a) \max_{a'} f(s', a')$$

- $\hat{\mathcal{P}}_{i,k}$ : empirical estimate of  $\mathcal{P}$  of agent  $i$  in epoch  $k$

Variance of  $\mathcal{T}_{i,k}$  down by  $H$  – less comm

# VRDQ – Module 2: Diffusion

- Agent  $i$  diffuses  $\mathcal{T}_{i,k-1} Q_{i,k}$  for  $L$  steps at the beginning of epoch  $k$ 
  - $Q_{i,k}$ : Q-estimate of agent  $i$  at the beginning of epoch  $k$
  - $\mathcal{T}_{i,k-1}$ : Agent  $i$  only has access to this refined operator at the beginning of epoch  $k$

- **Diffusion:**

$$d_{i,k}^{(0)} = \mathcal{T}_{i,k-1} Q_{i,k}, d_{i,k}^{(\ell+1)}(s, a) = \sum_{j \in \mathcal{N}_i} W_{ij} d_{i,k}^{(\ell)}(s, a), \ell = 0, \dots, L-1$$

- $W_{ij}$ : (i,j)-th entry of the weight matrix  $W$  associated with graph  $\mathcal{G}$
- Agent update:
- **Key observation:** the operator estimation time **dominates** the diffusion time – diffusion and estimation can evolve **in parallel!** – **no extra time for diffusion**

# Main results

**Theorem 1.** Under standard assumptions and a suitable choice of step-size, supposing each agent has access to  $T$  samples, then by selecting  $K = \left\lfloor \frac{c_1}{1-\gamma} \log NT \right\rfloor = \tilde{\mathcal{O}}(1)$ ,  $L = \left\lfloor \log c_2 N^{\frac{3}{2}} \sqrt{T(1-\gamma)} / \log 1/\rho \right\rfloor$ , our algorithm ensures w.p. at least  $1 - \delta$ :

$$\|Q_{i,k} - Q^*\|_\infty \leq \frac{\|Q_{i,0} - Q^*\|_\infty}{NT} + \tilde{\mathcal{O}}\left(\frac{\bar{R} \sqrt{\log NT \log \frac{2SAT}{\delta}}}{(1-\gamma)^{2.5} \sqrt{NT}}\right) = \tilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right)$$

## Takeaways:

- **Near-optimal** rate  $\tilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right)$  with **linear speedup**
- Only **Poly-Log communication (in both  $N, T$ )** required

# Simulation Results

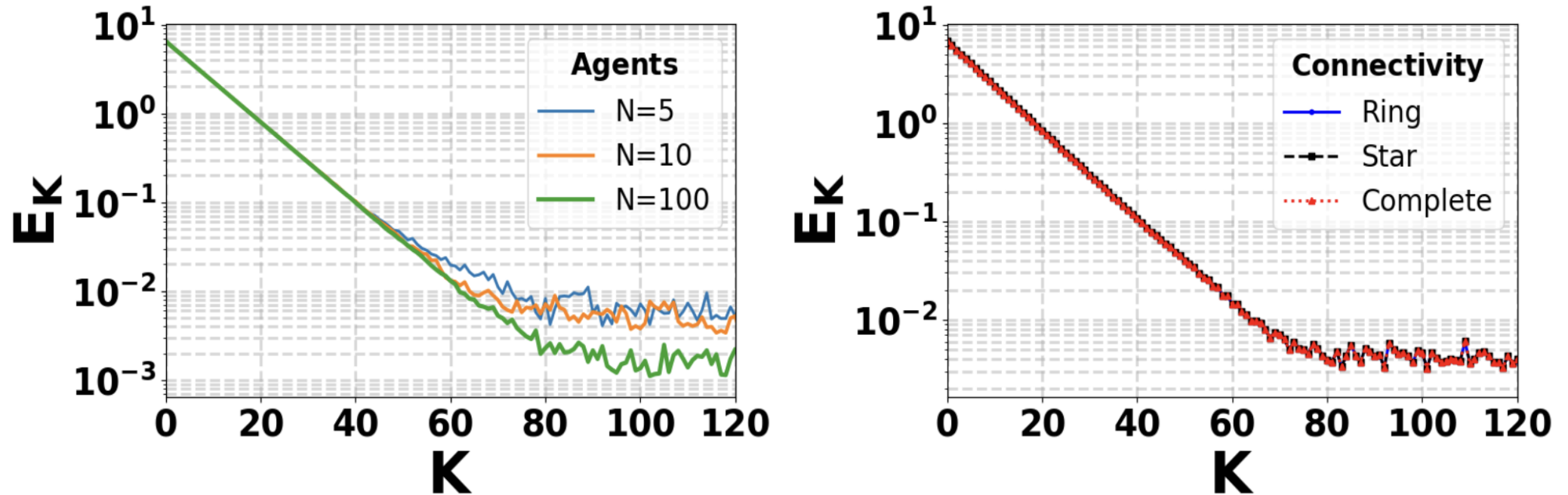


Fig. 2. Plots of the  $\ell_\infty$  error  $E_K = \sum_{i \in [N]} \|Q_{i,K} - Q^*\|_\infty / N$  for VRDQ as a function of the number of epochs  $K$ , with varying number of agents (**Left**), and under different network topologies with  $N = 100$  (**Right**).

# Extension to Time-Varying Networks

- Assume a standard joint strong connectivity condition [1]
- **Exact same** rate  $\tilde{O}\left(\frac{1}{\sqrt{NT}}\right)$  can be achieved
- Poly-log communication complexity preserved

# Conclusion

- Introduced a novel approach for decentralized Q-learning over static and time-varying networks
- Demonstrated near-optimal rates with collaborative speedups in sample-complexity
- Poly-logarithmic communication complexity overhead required
- Future: extension to function approximation and Markovian sampling

**Thank you!**  
**Questions?**

# Working Assumptions

- Bounded reward:  $R(s, a) \leq \bar{R}, \forall (s, a)$
- Doubly stochastic weighting matrix: Suppose  $W$  is the weighting matrix associated with the graph  $\mathcal{G}$ ; there exists  $C_1 > 0, \rho \in (0,1)$ , s.t. for all  $i, t$ :

$$\left\| [W^t]_i - \frac{1}{N} \mathbf{1}^\top \right\| \leq C_1 \rho^t$$

- Rows of  $W$  converge *exponentially* fast to the stationary distribution – yielding control over the diffusion error
- Suppose that there exists a *block parameter*  $B$  such that
$$\left\| W(t)W(t-1) \cdots W(t-B+1) - \frac{1}{N} \mathbf{1} \mathbf{1}^\top \right\| \leq \omega \in (0,1)$$